Baselines First

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One fair baseline-comparison record.

Fit from training, score aligned predictions, expose hidden errors and preserve the evaluation contract.

Keep one habit: W1–W3/W5 → selected independent comparison → project baseline record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

Dataset Cards

> **My remembered idea, example and correction**

Sampling and Intended Use

> **My remembered idea, example and correction**

Python and Table Readiness

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor training clear,clear,blocked,clear,blocked,clear chooses clear4vs2; tie chooses clear. ValidationV1–V4 blocked,clear,blocked,clear gives predictions clear×4,correctV2,V4=2/4. Supplied candidate blocked,clear,clear,clear correctV1,V2,V4=3/4,one more correct/25percentage points. Main baseline training4/6 is not validation accuracy.

  Separate leaf picture trainninghealthy,healthy,spotted;validationhealthy,spotted,healthy,spotted;candidatehealthy,spotted,spotted,spotted:2/4vs3/4,repairthirdcandidate→4/4practiceonly. Equal scores need not mean equal errors.

- Reference for this exercise: open this week's student lesson, "Baselines First", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Count clear and blocked in the worked **training** list. State the constant **prediction** and the declared tie rule. Explain why the validation majority does not choose this **baseline**.

My reasoning and evidence

W2 · Trace the current experiment

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor training clear,clear,blocked,clear,blocked,clear chooses clear4vs2; tie chooses clear. ValidationV1–V4 blocked,clear,blocked,clear gives predictions clear×4,correctV2,V4=2/4. Supplied candidate blocked,clear,clear,clear correctV1,V2,V4=3/4,one more correct/25percentage points. Main baseline training4/6 is not validation accuracy.

  Separate leaf picture traininghealthy,healthy,spotted;validationhealthy,spotted,healthy,spotted;candidatehealthy,spotted,spotted,spotted:2/4vs3/4,repairthirdcandidate→4/4practiceonly. Equal scores need not mean equal errors.

- List the four **baseline prediction**s and mark which validation rows are correct. Give correct/total and **accuracy**. Explain why 4/6 is not the answer.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor training clear,clear,blocked,clear,blocked,clear chooses clear4vs2; tie chooses clear. ValidationV1–V4 blocked,clear,blocked,clear gives predictions clear×4,correctV2,V4=2/4. Supplied candidate blocked,clear,clear,clear correctV1,V2,V4=3/4,one more correct/25percentage points. Main baseline training4/6 is not validation accuracy.

  Separate leaf picture traininghealthy,healthy,spotted;validationhealthy,spotted,healthy,spotted;candidatehealthy,spotted,spotted,spotted:2/4vs3/4,repairthirdcandidate→4/4practiceonly. Equal scores need not mean equal errors.

- A candidate is scored on ten easy rows and the **baseline** on four difficult rows. Describe a fair comparison and two details you must record.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main corridor training clear,clear,blocked,clear,blocked,clear chooses clear4vs2; tie chooses clear. ValidationV1–V4 blocked,clear,blocked,clear gives predictions clear×4,correctV2,V4=2/4. Supplied candidate blocked,clear,clear,clear correctV1,V2,V4=3/4,one more correct/25percentage points. Main baseline training4/6 is not validation accuracy.

  Separate leaf picture traininghealthy,healthy,spotted;validationhealthy,spotted,healthy,spotted;candidatehealthy,spotted,spotted,spotted:2/4vs3/4,repairthirdcandidate→4/4practiceonly. Equal scores need not mean equal errors.

- A **training**-derived clear **baseline** gets nine of ten validation cases correct; the labels are nine clear and one blocked. State its **accuracy** and missed case. If a candidate also gets nine correct, must it miss the same case?

  Explain.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Keep this task's new training labels and validation order separate from the worked corridor and pictured leaf cases. Use the same majority rule with the predeclared clear-on-tie policy. Candidate predictions are supplied; no candidate fitting or future-performance result is provided.

- New **training** labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same **baseline** rule and list its four **prediction**s.

  A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and **percentage point**s. What claim remains unsupported?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Fit and compare independently

> **My independent explanation, trace and limits**
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> |---|
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MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Record majority and transparent-**rule baseline**s for the Plant Classifier using the frozen test plan.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Calculate the validation score for A **baseline** before a model. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for A **baseline** before a model. Proposed explanation: “A simple **baseline** does not need **training** data”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- A dataset contains 27 healthy and 3 spotted leaves. An always-healthy **baseline** gets what **accuracy** and spotted recall? What must a new model improve upon?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A test set has eighteen healthy and two spotted leaves. **Baseline** always predicts healthy.
- Calculate and critique **accuracy**.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A dataset has 18 grass and 2 fern examples. A **baseline** always predicts grass. Compute **accuracy** and name a serious limitation.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 10 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A dataset contains 27 healthy and 3 spotted leaves. An always-healthy **baseline** gets what **accuracy** and spotted recall? What must a new model improve upon?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- New **training** labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same **baseline** rule and list its four **prediction**s.

  A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and **percentage point**s. What claim remains unsupported?

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Make a simple **baseline** earn its place before claiming a better classifier.
- **Training** labels: healthy, healthy, spotted. Majority **baseline** predicts healthy. Validation references: healthy, spotted, healthy, spotted.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/10

Baselines First

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit baseline

Training majority is healthy: 2 of 3 labels.



2. Score baseline

Always healthy matches 2/4 validation cases: 50%.



3. Compare candidate

Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

CHANGE ONE THING

Change the third candidate prediction from spotted to healthy.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Make a simple **baseline** earn its place before claiming a better classifier.
- Which labelled step matches this detail? Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.
- 1. Fit **baseline**
- 2. Score **baseline**
- 3. Compare candidate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> | |
> |---|
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▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare candidate Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. Read the picture in order.

Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Change the third candidate **prediction** from spotted to healthy. Which conclusion is supported by the supplied facts?
- 1. Candidate becomes correct on all 4 validation cases.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Candidate becomes correct on all 4 validation cases. Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Repair and keep the record

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sampling and Intended Use (ai-4/7). Goal: Define the population and conditions in an intended-use statement. Short practice: Which collection would address the intended-use gap most directly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Python and Table Readiness (ai-4/4). Goal: Specify a table schema. Short practice: Why should the missing value not silently become zero?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

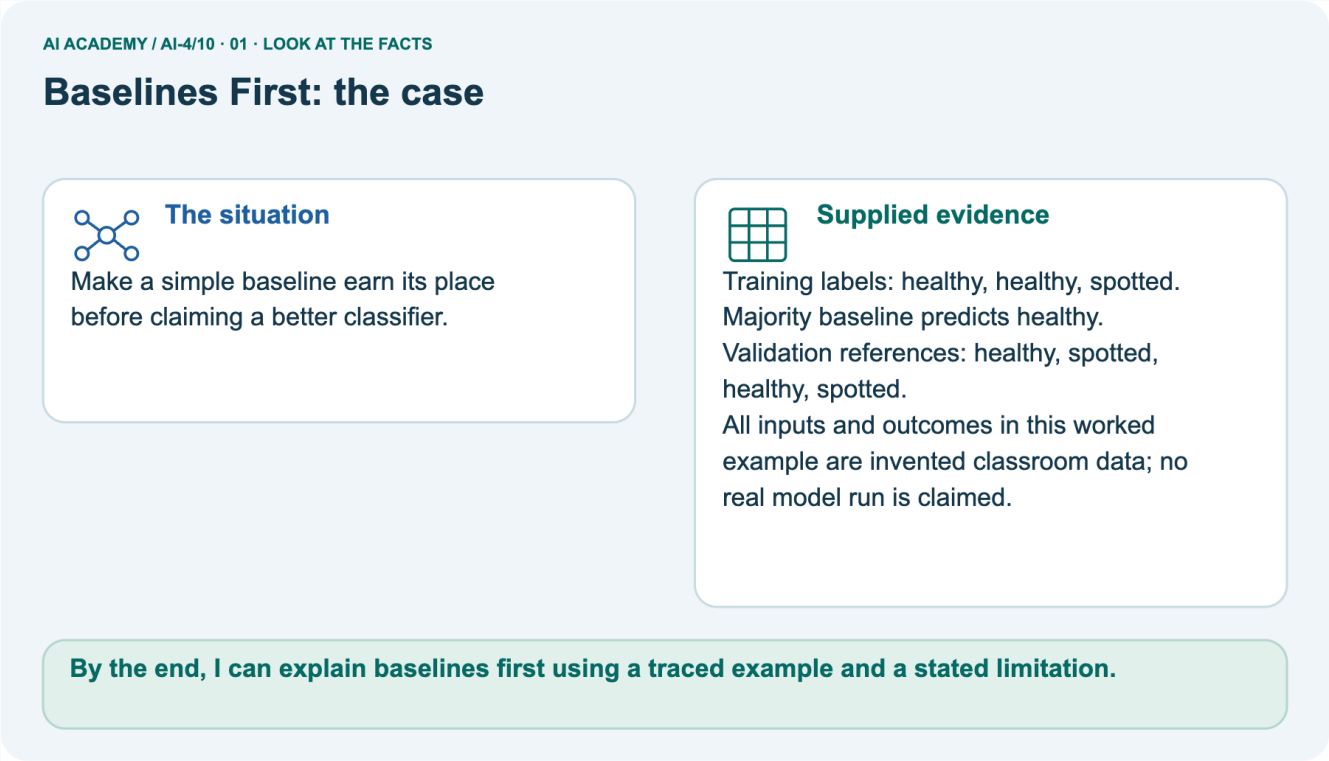
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the relevant picture instead of repeating every task.

01. Baselines First: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

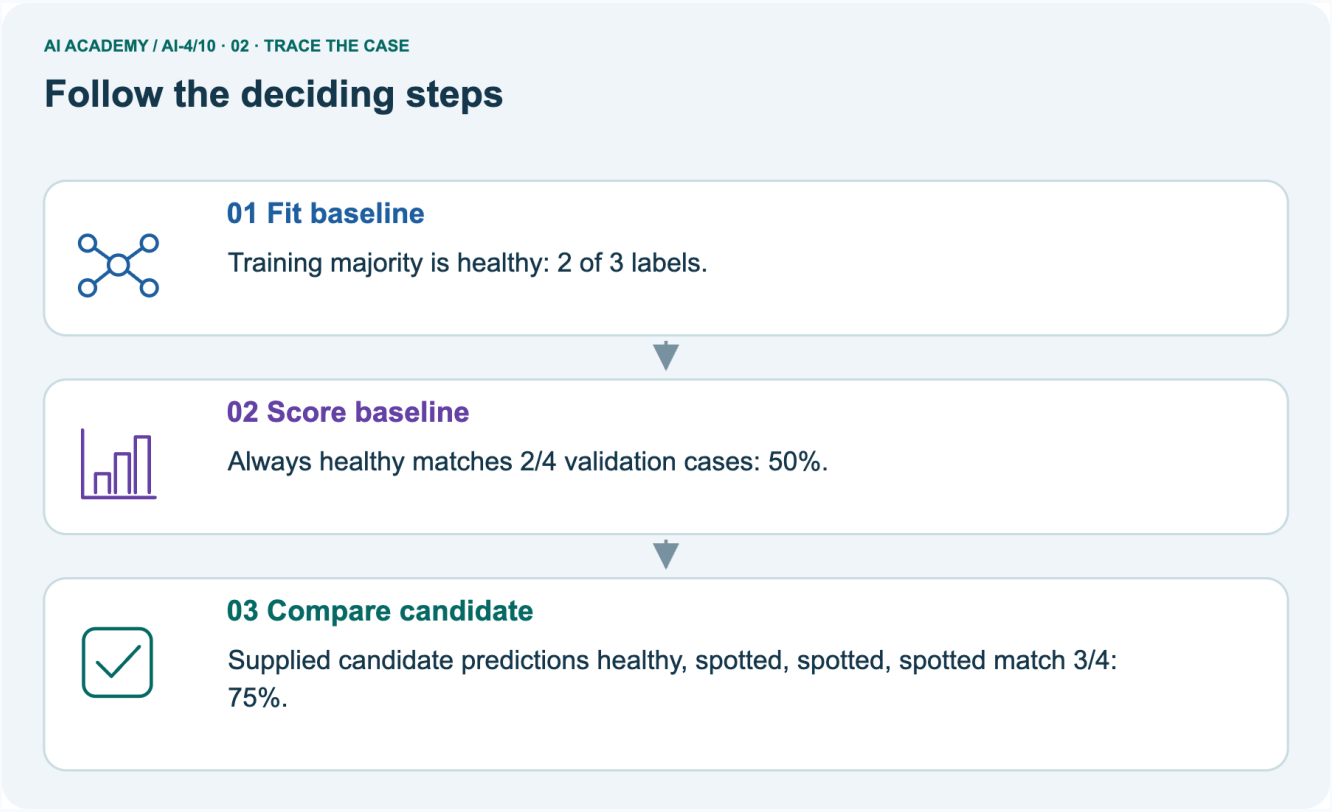
Make a simple **baseline** earn its place before claiming a better classifier. Supplied facts: **Training** labels: healthy, healthy, spotted. Majority **baseline** predicts healthy.

Validation references: healthy, spotted, healthy, spotted. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



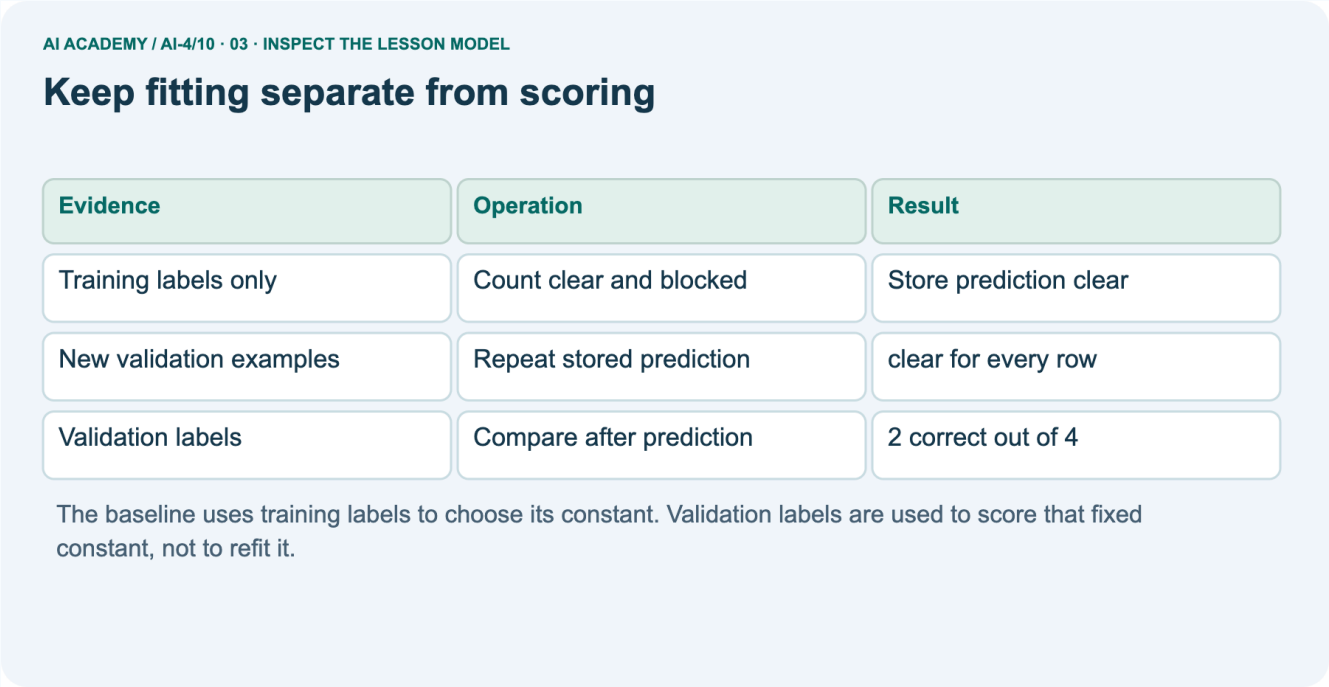
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **baseline**: **Training** majority is healthy: 2 of 3 labels. → Score **baseline**: Always healthy matches 2/4 validation cases: 50%. → Compare candidate: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep fitting separate from scoring



Follow the labelled picture with the matching explanation. Open the vector diagram

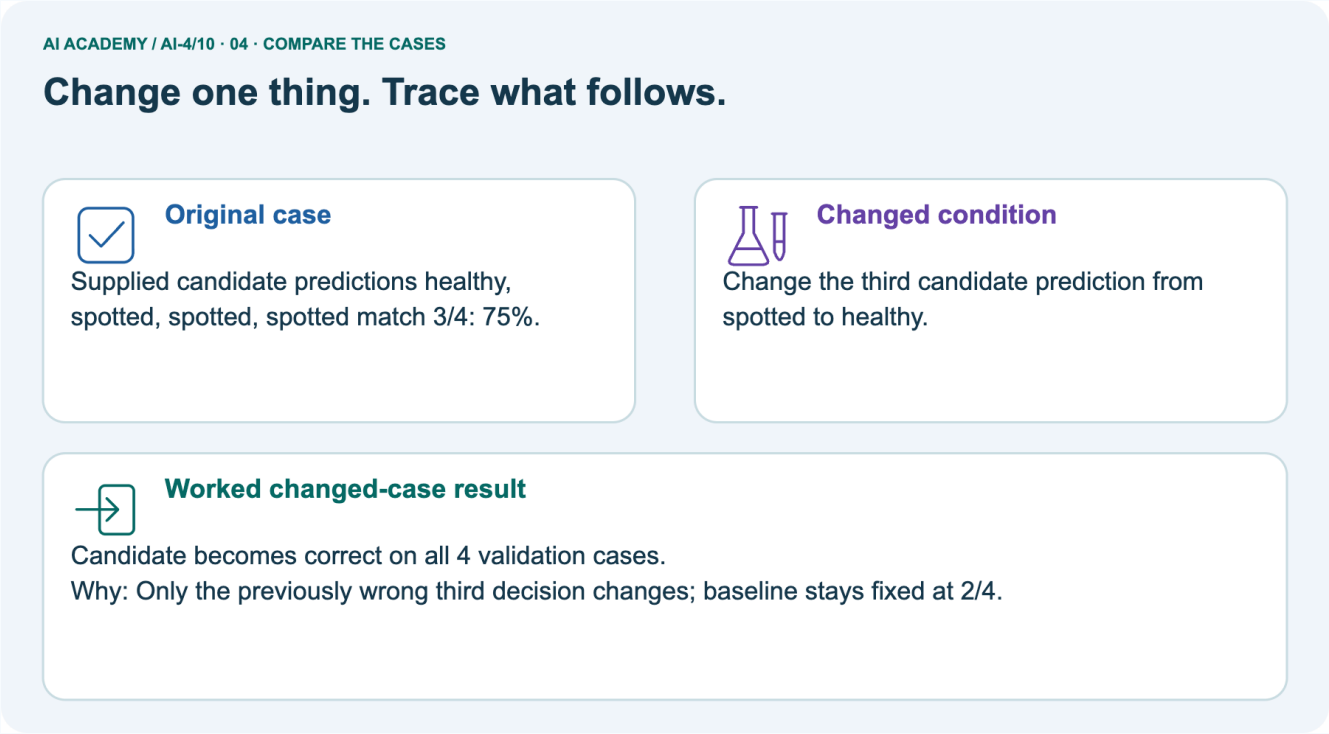
Keep **fitting** separate from scoring. **Training** labels only | Count clear and blocked | Store **prediction** clear. New validation examples | Repeat stored **prediction** | clear for every row.

Validation labels | Compare after **prediction** | 2 correct out of 4

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

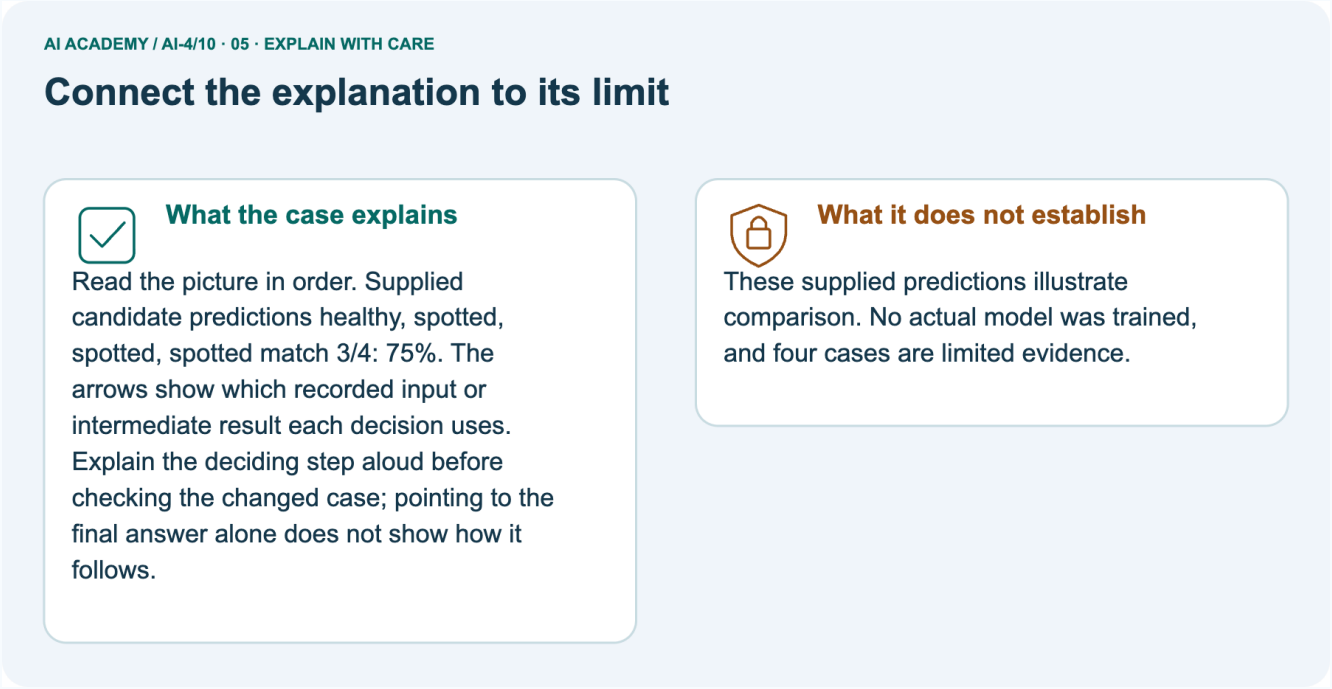
Original case: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. Changed condition: Change the third candidate **prediction** from spotted to healthy.

Expected result: Candidate becomes correct on all 4 validation cases. Reason: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

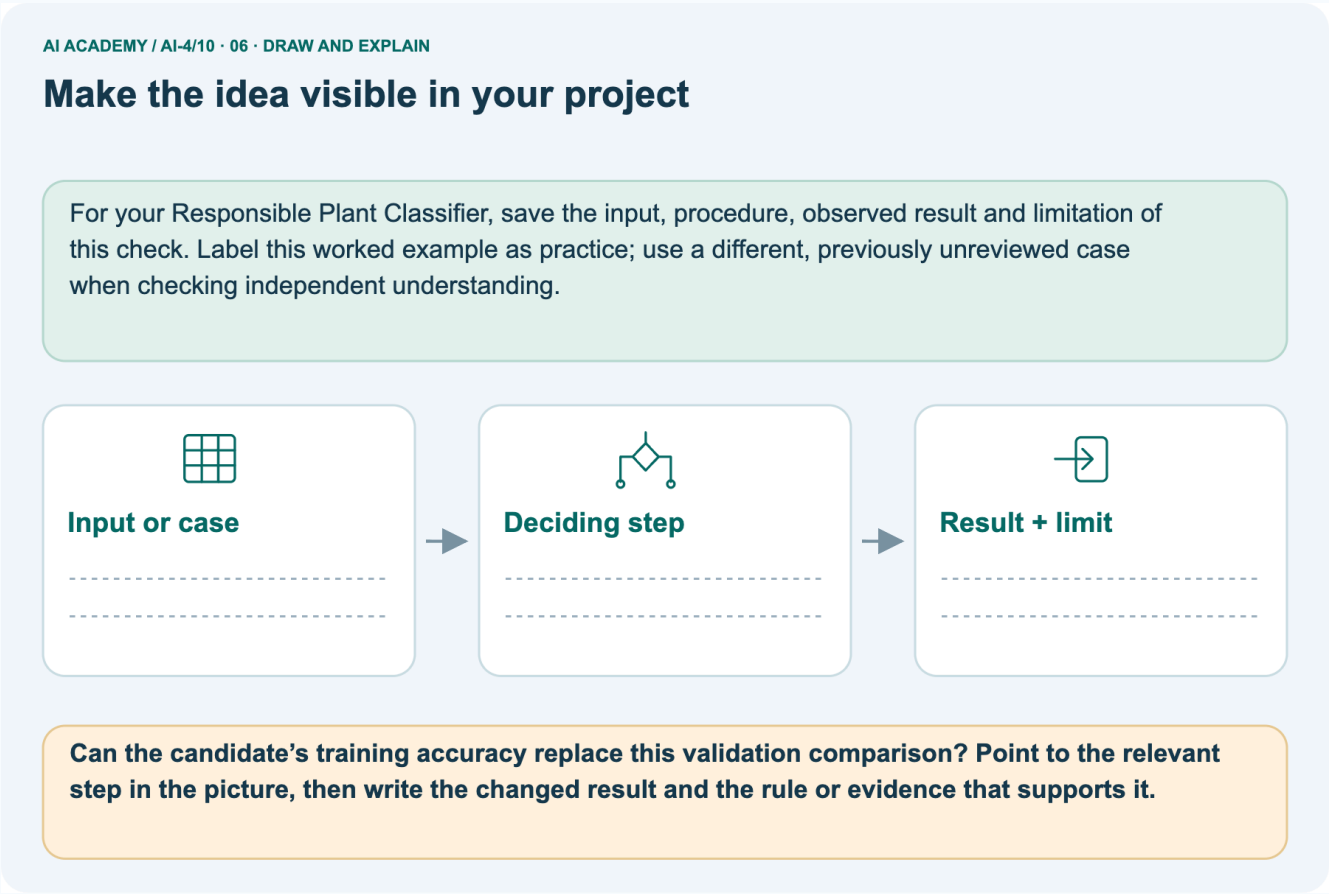
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: These supplied **prediction**s illustrate comparison.

No actual model was trained, and four cases are limited **evidence**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Baselines First: the case

02. Follow the deciding steps

03. Keep fitting separate from scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.